

Cross-Border IPS AI Agent

Federated, FHIR-native terminology alignment for cross-border Type 2 diabetes summaries.

FHIR IPS is the interoperable artifact.
PDFs are human-readable renderings.

Working demo · synthetic evidence · official IPS 2.0.1 validation

IPS AI Agent

Federated terminology + FHIR R4 IPS-style document Bundles

4 sites

IPS 2.0.1: 0 errors

step 01

down (default)

Target

IND

Summary / source PDF

ing document. HL7 FHIR m-facing interoperability

sd USA source PDF

ype 2 diabetes

year old male with

ood pressure has

creatinine 1.1

metformin 1000 mg

.0 mg.

it used by the validated

un pipeline

try

the local site.

ises stay local.

tay local.

undles stay local.

s model tensors + sample

CURRENT EXCHANGE ROUTE

United States > India

semantic interoperability

FHIR document exchange

ABDM

WHY HL7 MATTERS HERE

Country PDFs are readable documents; HL7 FHIR IPS is the interoperable source of truth.

The demo converts local clinical wording into standard codes, packages them in FHIR IPS, then renders a receiver-friendly country report from that Bundle.

HL7-centered exchange

SOURCE COUNTRY PDF

US Patient Care Summary / C-CDA-style source PDF

USA human-readable clinical input

HL7 INTEROPERABILITY LAYER

HL7 FHIR IPS document Bundle

canonical machine-readable source of truth for exchange

TARGET COUNTRY PDF

Final Indian receiver

ABDM FHIR R4 readiness via rendering

CASE MAPPING COVERAGE

100%

6/6 facts · registry + FL

FHIR BUNDLE TYPE

document

IPS-style FHIR R4

OFFICIAL IPS VALIDATION

0 errors

0 warnings · 4/4 Bundles

SYNTHETIC EVIDENCE

48 / 4

48 received

01 Report

02 Facts

03 Terminology

04 FL Linker

05 ConceptMap

06 IPS Bundle

End-to-end clinical QA path

What makes the Bundle medically reviewable, not just structurally valid.

SOURCE-TO-FACT

6 retained facts

source phrases stay visible for audit

FACT-TO-CODE

SNOMED CT + LOINC + RxNorm + ICD-10

registry first, federated linker on misses

CODE-TO-FHIR

Condition / Observation / MedicationStatement

clinical facts placed into expected resources

DOCUMENT P

Composition first

Bundle.type=document

document id

Clinical Trace

FHIR Terminology

Federated Learning

FHIR IPS Bundle

Evidence & Limitations

Clinical fact extraction → terminology binding

Rule-backed extractor (prototype). Pretrained clinical NER is future work.

6 facts

SNOMED CT · LOINC

PHRASE	NORMALIZED CONCEPT	CODE(S)	FHIR RESOURCE
T2DM	Type 2 diabetes mellitus	SNOMED CT 44054006 ICD-10 E11	Condition

One condition arrives under many local names

USA	T2DM
India	sugar disease · madhumeha type 2
Australia	type 2 DM
Europe	DM2 · diabetes mellitus type II



Canonical meaning

Type 2 diabetes mellitus

SNOMED CT 44054006

ICD-10 E11

Without semantic alignment, a receiver EHR gets ambiguous text instead of computable clinical data.

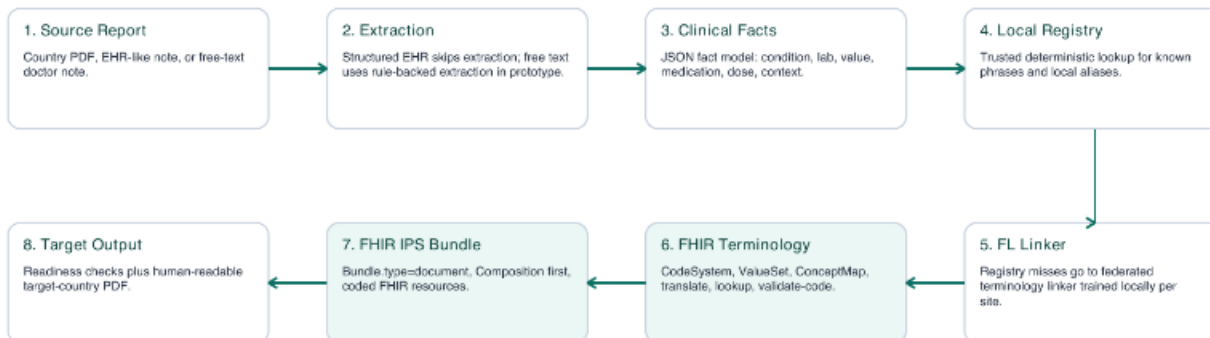
[illegible]

Every transformation remains inspectable

The registry is deterministic; the federated linker is used only on misses; FHIR terminology validates and publishes the mapping.

Cross-Border IPS AI Agent - Architecture

FHIR IPS is the interoperable artifact; PDFs are human-readable renderings.



Interoperability and privacy evidence

Semantic interoperability: SNOMED CT, ICD-10, LOINC, RxNorm, and FHIR ConceptMap.

Data interoperability: Patient, Condition, Observation, MedicationStatement, Composition, and Bundle.

Federated privacy boundary: raw reports, labels, aliases, identifiers, and patient-level Bundles stay local.

Coordinator receives model tensors and sample counts only. FedAvg is data locality, not formal privacy.

Five seeds reproduce the cross-site gain without moving reports

48/48

federated transfer at all 5 seeds

47/48

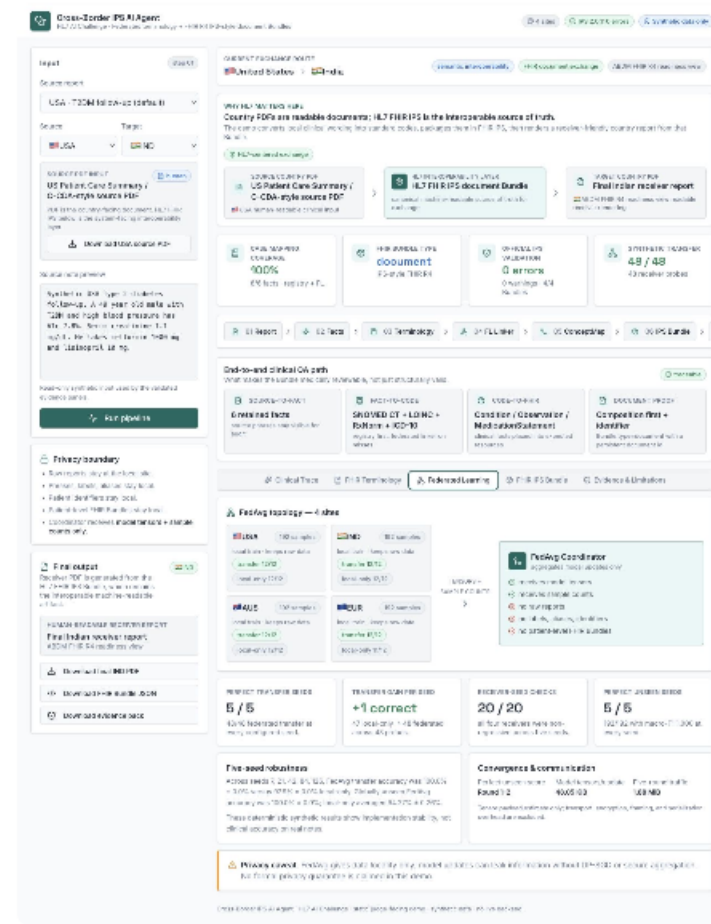
local-only transfer at all 5 seeds

+1

measured gain per seed

20/20 receiver-seed checks were non-regressive

1.88 MiB five-round tensor traffic. Synthetic study; not clinical accuracy.



FHIR turns an AI prediction into an auditable mapping

The interface displays a pipeline for processing FHIR data. It includes a 'Run pipeline' button, a 'Privacy boundary' section with rules, and a 'Final output' section with download options. The main area shows three panels: 'CodeSystem - local site terms', 'ValueSet - allowed target concepts', and 'ConceptMap - local phrase → standard terminology'. Each panel contains JSON data representing FHIR resources.

CodeSystem - local site terms

```
{
  "resourceType": "CodeSystem",
  "id": "local-usa-diabetes-terms",
  "url": "https://ips-agent.deno/CodeSystem/local-usa-diabetes-terms",
  "version": "0.1.0",
  "name": "LocalUSADiabetesTerms",
  "status": "active",
  "content": "complete",
  "concept": [
    {
      "code": "T2DM",
      "display": "Type 2 diabetes mellitus (local abbrev.)"
    }
  ]
}
```

ValueSet - allowed target concepts

```
{
  "resourceType": "ValueSet",
  "id": "ips-diabetes-target-concepts",
  "url": "https://ips-agent.deno/ValueSet/ips-diabetes-target-concepts",
  "status": "active",
  "compose": {
    "include": [
      {
        "system": "http://snomed.info/sct",
        "concept": [
          {
            "code": "44054006",
            "display": "Diabetes mellitus type 2"
          }
        ]
      }
    ]
  }
}
```

ConceptMap - local phrase → standard terminology

```
{
  "resourceType": "ConceptMap",
  "id": "local-usa-to-ips-diabetes",
  "url": "https://ips-agent.deno/ConceptMap/local-usa-to-ips-diabetes",
  "status": "active",
  "sourceUri": "https://ips-agent.deno/CodeSystem/local-usa-diabetes-terms",
  "group": [
    {
      "source": "https://ips-agent.deno/CodeSystem/local-usa-diabetes-terms",
      "target": "http://snomed.info/sct",
      "element": [
        {
          "code": "T2DM",
          "display": "Type 2 diabetes mellitus (local)",
          "target": [
            {
              "code": "44054006",
              "display": "Diabetes mellitus type 2"
            }
          ]
        }
      ]
    }
  ]
}
```

1 **CodeSystem**
represents local site terms

2 **ValueSet**
constrains allowed targets

3 **ConceptMap + local \$translate**
publishes local → standard

4 **External \$lookup / \$validate-code**

tx.fhir.org accepted 4/4 representative codes

External code validity does not prove phrase mapping; low confidence → human review

All four exchange Bundles pass the IPS 2.0.1 profile

4/4

current route Bundles

0

errors

0

warnings

2 information notes per Bundle

RxNorm ingredient codes are outside the IPS guide's recommended medication ValueSet.

Read-only synthetic input used by the validated evidence panels.

Run pipeline

Privacy boundary

- Raw reports stay at the local site.
- Phrases, labels, aliases stay local.
- Patient identifiers stay local.
- Patient-level FHIR Bundles stay local.
- Coordinator receives model tensors + sample counts only.

Final output

Receiver PDF is generated from the HL7 FHIR IPS Bundle, which remains the interoperable machine-readable artifact.

HUMAN-READABLE RECEIVER REPORT

Final Indian receiver report

ABDM FHIR R4 readiness view

Download final IND PDF

Download FHIR Bundle JSON

Download evidence pack

SOURCE-TO-FACT

6 retained facts

source phrases stay visible for audit

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DOCUMENT PROOF

Composition first + identifier

Bundle type=document with a persistent document id

Clinical Trace FHIR Terminology Federated Learning **FHIR IPS Bundle** Evidence & Limitations

International Patient Summary — Synthetic USA T2DM

Composition-first FHIR R4 document Bundle - USA → IND (ABDM FHIR R4 readiness view). All four current Bundles pass the official HL7 validator against IPS 2.0.1 with 0 errors and 0 warnings.

Bundle type = document Identifier present 9 entries

Final Indian receiver report

Human-readable receiver view generated from the FHIR IPS Bundle. The Bundle remains the source of truth for system exchange.

USA source IND receiver Download final IND PDF Bundle JSON

PATIENT

Synthetic patient

male · DOB 1977-04-12

RECEIVER FORMAT

ABDM FHIR R4 readiness view

readiness-only rendering, not certification

SOURCE DOCUMENT

ips-bundle-demo-usa-001

FHIR Bundle.type=document

PROBLEMS

Diabetes mellitus type 2

reported condition

SNOMED CT 44054006 ICD-10 E11

Hypertensive disorder

reported condition

SNOMED CT 38341003 ICD-10 I10

RESULTS

Hemoglobin A1c/Hemoglobin.total in Blood

7.8 %

LOINC 4548-4

Creatinine [Mass/volume] in Serum or Plasma

1.1 mg/dL

LOINC 2180-0

MEDICATIONS

metformin

1000 mg

RxNorm 6809

lisinopril

10 mg

RxNorm 29046

RECEIVER HANDOVER NOTES

- ABDM FHIR R4-oriented readiness view; ABHA identifier is flagged because no real national identifier exists for synthetic data.
- The receiving Indian system can store accepted ConceptMap mappings in its local learned cache for reuse.

RESOURCES

Composition	composition-1
Patient	patient-1
Condition	cond-42dm
Condition	cond-htn

Composition

```

{
  "resourceType": "Composition",
  "id": "composition-1",
  "status": "final",
  "type": (

```

Structural/profile conformance for synthetic Bundles; not clinical or national certification.

One machine-readable summary supports four receiver journeys

USA → India · India → USA · Australia → Europe · Europe → USA

The image displays eight PDF thumbnails arranged in a 2x4 grid, illustrating the machine-readable summary for four different receiver journeys. Each thumbnail shows a structured report with a header, patient information, clinical history, and a table of findings.

- Original US doctor report** (Top Left): Shows a report from a US doctor to an Indian receiver. The report includes patient information, clinical history, and a table of findings.
- Final Indian receiver report** (Top Second): Shows the final report received by the Indian receiver, which is a machine-readable summary of the original report.
- Original Indian doctor report** (Top Third): Shows a report from an Indian doctor to a US receiver. The report includes patient information, clinical history, and a table of findings.
- Final US receiver report** (Top Fourth): Shows the final report received by the US receiver, which is a machine-readable summary of the original report.
- Original Australian doctor report** (Bottom Left): Shows a report from an Australian doctor to a European receiver. The report includes patient information, clinical history, and a table of findings.
- Final European receiver report** (Bottom Second): Shows the final report received by the European receiver, which is a machine-readable summary of the original report.
- Original European doctor report** (Bottom Third): Shows a report from a European doctor to a US receiver. The report includes patient information, clinical history, and a table of findings.
- Final US receiver report** (Bottom Fourth): Shows the final report received by the US receiver, which is a machine-readable summary of the original report.

The PDFs are readable renderings. The FHIR IPS Bundle remains the system-facing source of truth.

The evidence is executable, downloadable, and reviewable

16/16

frontend behavior and evidence tests

13/13

Python pipeline and FedAvg tests

16/16

route downloads click-tested

```
$ npm test -- --run  
✓ 16 tests passed
```

```
$ PYTHONPATH=src pytest -q  
✓ 13 tests passed
```

```
$ HL7 FHIR Validator + IPS 2.0.1  
✓ 4 Bundles · 0 errors · 0 warnings
```

```
$ tx.fhir.org terminology snapshot  
✓ 4/4 lookup + validate-code
```

GitHub includes the PyTorch clients, FedAvg coordinator, external terminology snapshot, Bundles, review packet, and unedited validator logs.

The standards do the traveling. The reports stay local.

Federated learning

aligns local clinical terminology
without centralizing raw reports

FHIR terminology

makes CodeSystem, ValueSet,
ConceptMap, and validation
evidence auditable

FHIR IPS

packages coded facts into a
cross-border patient summary

Working demo

crossroad-fhir-link-three.vercel.app